SMART-BIN

AN ESP32-BASED IoT WASTE BIN MONITORING SYSTEM WITH FILL, WEIGHT, AIR-QUALITY AND TAMPER DETECTION FOR PRIORITY-BASED COLLECTION

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SDG 11 Sustainable Cities and Communities

THE PROBLEM

60.76K tons/day

of garbage in the Philippines, 2025 estimate (PSA, 2026)

Metro Manila alone: **10,368 MT a day** projected by 2025 (World Bank, 2022)

Fixed collection schedules: some bins overflow before pickup, others get emptied half full.

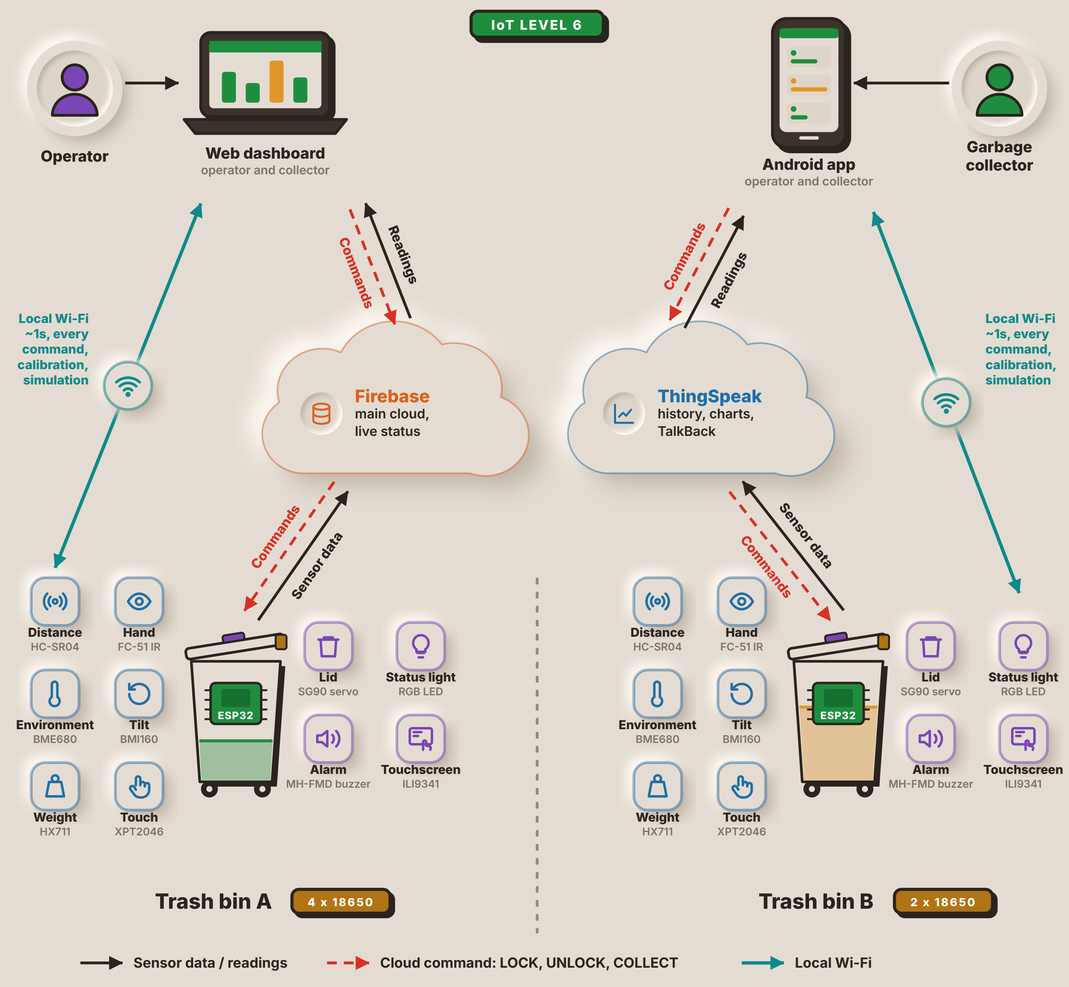
WHAT SMART-BIN DOES

01 · SENSE
Sensor monitoring
Fill, weight, temperature, humidity, air quality, tilt and tamper on two bins. The lid opens for a hand.

02 · ALERT
Real-time alerts
Live data and alerts on a web dashboard, an Android app and the bin's touchscreen.

03 · PRIORITIZE
Collection priority
Bins ranked by fullness and time to full. Remote lock, PIN collection and calibration.

HOW IT WORKS

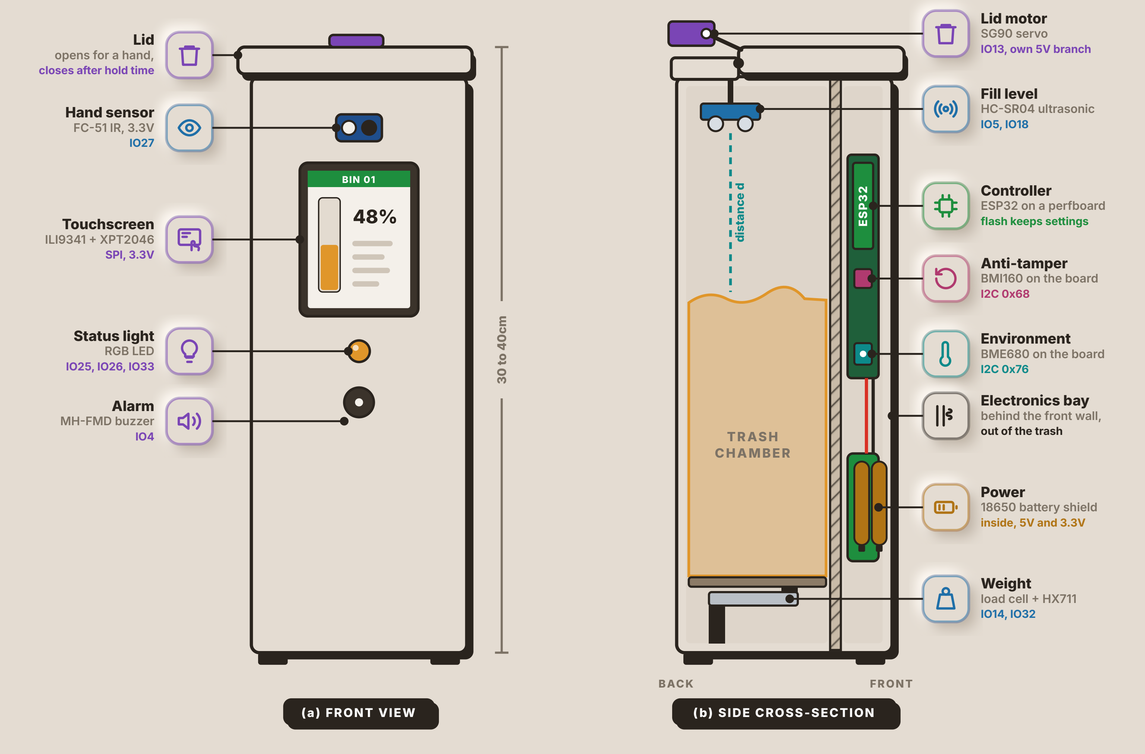


Firebase keeps the live status, ThingSpeak keeps the history. On the same Wi-Fi the apps reach a bin directly. IoT Level 6 (Bahga & Madisetti, 2014).

THE PROTOTYPE

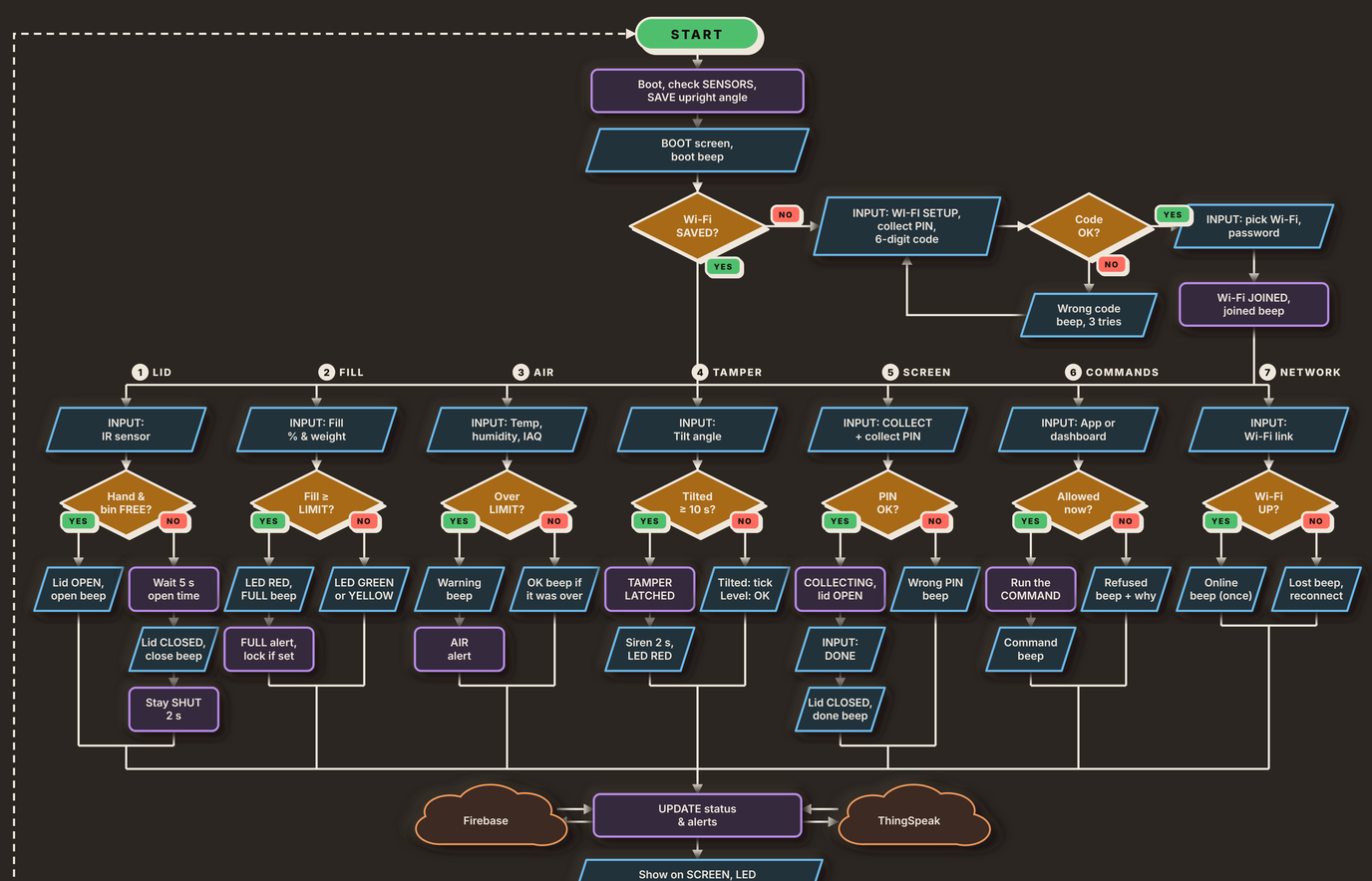
ONE BIN UNIT, BUILT TWICE (BIN A AND BIN B)

Bin A runs on a 4 × 18650 shield, Bin B on a 2 × 18650 shield; every other part is the same.



2 bins built	6 sensors per bin	4 outputs per bin	₱5,955 parts, both bins

SYSTEM FLOWCHART



THE BUILD



DASHBOARD, APP AND TOUCHSCREEN

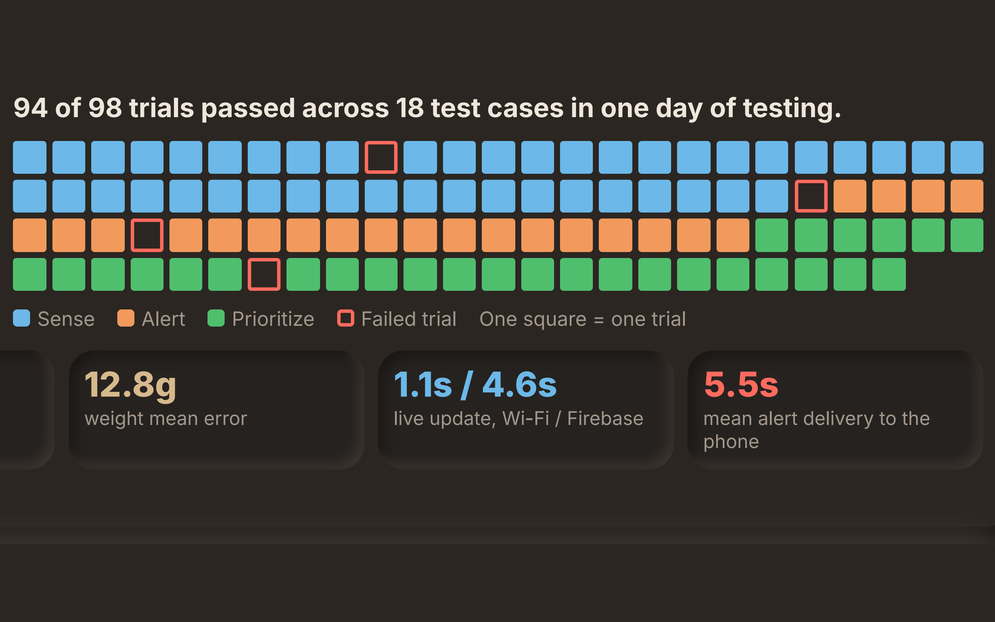


- Priority list
- 5 alert types
- Remote lock
- PIN collection
- Time to full
- Calibration

RESULTS

95.9% TRIALS PASSED

94 of 98 trials passed across 18 test cases in one day of testing.



1.8 pts	12.8g	1.1s / 4.6s	5.5s
fill level mean error	weight mean error	live update, Wi-Fi / Firebase	mean alert delivery to the phone

CONCLUSION

SENSE · MET
Fill within 1.8 points and weight within 12.8g on average. The tamper alarm and the lid acted on time.

ALERT · MET
New readings in under 2s on local Wi-Fi and under 8s through Firebase. Alerts reached the phone in 5.5s.

PRIORITIZE · MET
Right order in all 10 priority tests. Time to full is a guide for ordering bins, not an exact time.

Smart-Bin monitors the fill, weight and air of each bin, detects tampering, and tells the operator and the collector which bin to empty first.

SMART-BIN

Adviser: **Prof. Jonel R. Macalisang** Code, app and paper: **github.com/GenReyes/Smart-Bin-Embedded-Systems**

Bahga, A., & Madisetti, V. (2014). *Internet of things: A hands-on approach.* · Philippine Statistics Authority. (2026). *Highlights of the Compendium of Philippine Environment Statistics 2016–2025, Component 3: Residuals.* · World Bank. (2022). *An assessment of municipal solid waste plans, collection, recycling and disposal of Metro Manila.*